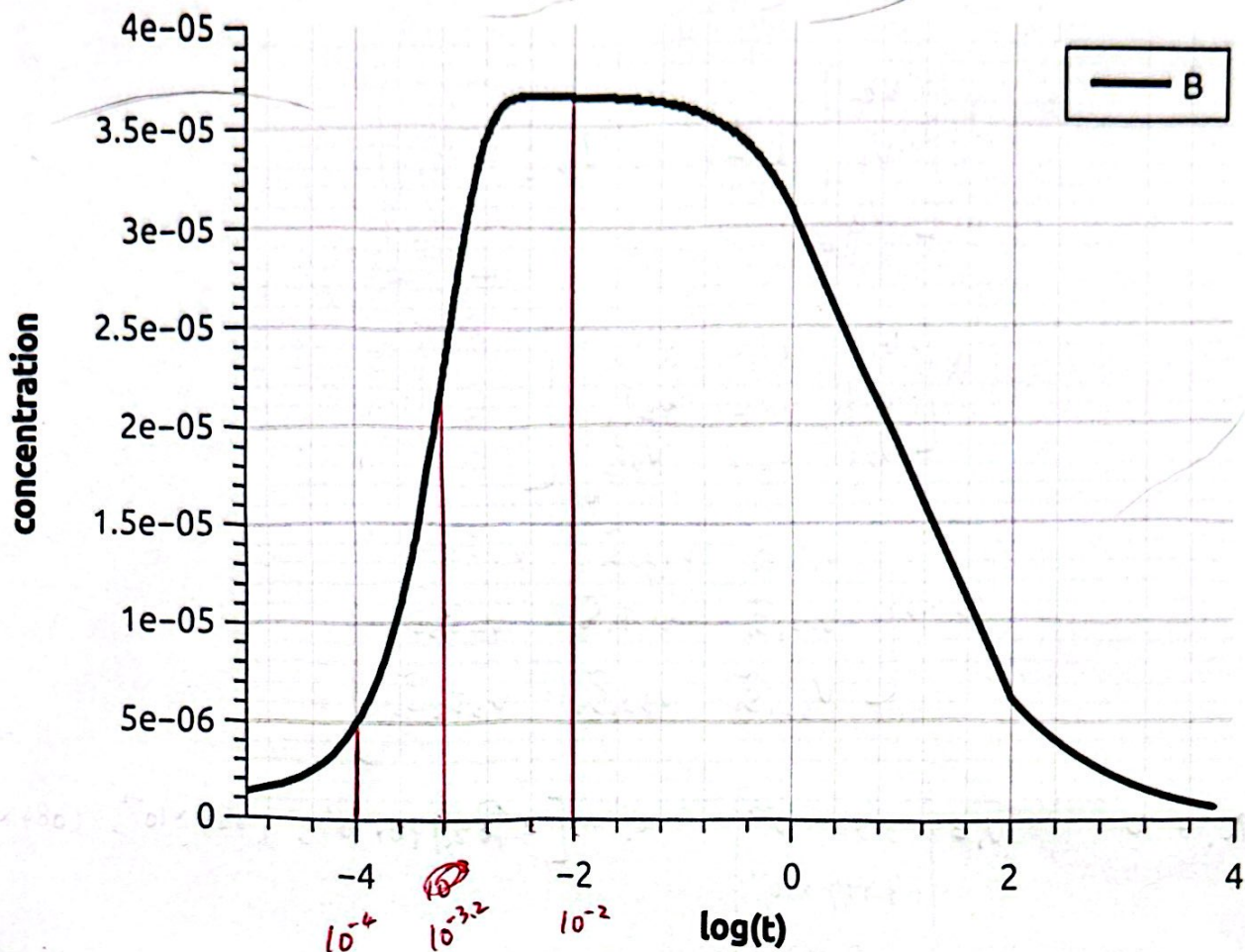


Concentration of species B vs log t



①

$$t = 10^{-3.2}, \quad \lambda_1 = 0, \quad V_1 = \begin{bmatrix} 2.655 \times 10^{-5} \\ 1.000 \\ 0.000 \end{bmatrix}, \quad V_1^{-1} = \begin{bmatrix} 2.039 \times 10^4 & 4.387 \times 10^{-1} & 9.623 \times 10^{-2} \end{bmatrix}$$

$$\lambda_2 = -8.7 \times 10^{-2}, \quad V_2 = \begin{bmatrix} -2.250 \times 10^{-5} \\ 2.000 \\ 0.000 \end{bmatrix}, \quad V_2^{-1} = \begin{bmatrix} -2.039 \times 10^4 & 5.413 \times 10^{-1} & 1.135 \times 10^{-1} \end{bmatrix}$$

$$\lambda_3 = 3.218 \times 10^{-3}$$

$$V_3 = \begin{bmatrix} -6.776 \times 10^{-1} \\ -2.053 \times 10^{-1} \\ 9.787 \times 10^{-1} \end{bmatrix}$$

$$V_3^{-1} = \begin{bmatrix} 0 & 0 & 1.0218 \end{bmatrix}$$

concentration of A does not change that much.

rec1, rec2, don't take part in this process.

each eigenvector:

$$\lambda \underline{V} \underline{V}^{-1} \begin{bmatrix} \tilde{s}_1 \\ \tilde{s}_2 \\ \tilde{s}_3 \end{bmatrix}$$

$$= \lambda \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} [I_1 \ I_2 \ I_3] \begin{bmatrix} \tilde{s}_1 \\ \tilde{s}_2 \\ \tilde{s}_3 \end{bmatrix}$$

$$= \lambda \begin{bmatrix} V_1 I_1 & V_1 I_2 & V_1 I_3 \\ V_2 I_1 & V_2 I_2 & V_2 I_3 \\ V_3 I_1 & V_3 I_2 & V_3 I_3 \end{bmatrix} \begin{bmatrix} \tilde{s}_1 \\ \tilde{s}_2 \\ \tilde{s}_3 \end{bmatrix}$$

$$= \lambda \begin{bmatrix} V_1 I_1 \tilde{s}_1 & V_1 I_2 \tilde{s}_1 & V_1 I_3 \tilde{s}_1 \\ V_2 I_1 \tilde{s}_1 & V_2 I_2 \tilde{s}_1 & V_2 I_3 \tilde{s}_1 \\ V_3 I_1 \tilde{s}_1 & V_3 I_2 \tilde{s}_1 & V_3 I_3 \tilde{s}_1 \end{bmatrix}$$

②

$t = 10^{-2}$ $\lambda_1 = 0$
0.0.1.

$$V_1 = \begin{bmatrix} 3.331 \times 10^{-3} \\ 9.999 \times 10^{-1} \\ 0.000 \end{bmatrix}$$

$$V_1^{-1} = [2.970 \times 10^2, 1.084 \times 10^{-2}, 1.084 \times 10^{-4}]$$

$\lambda_2 = -3.687$

$$V_2 = \begin{bmatrix} -3.65 \times 10^{-5} \\ 0.999 \\ 0.000 \end{bmatrix}$$

$$V_2^{-1} = [-2.970 \times 10^2, 9.892 \times 10^{-1}, 9.566 \times 10^{-4}]$$

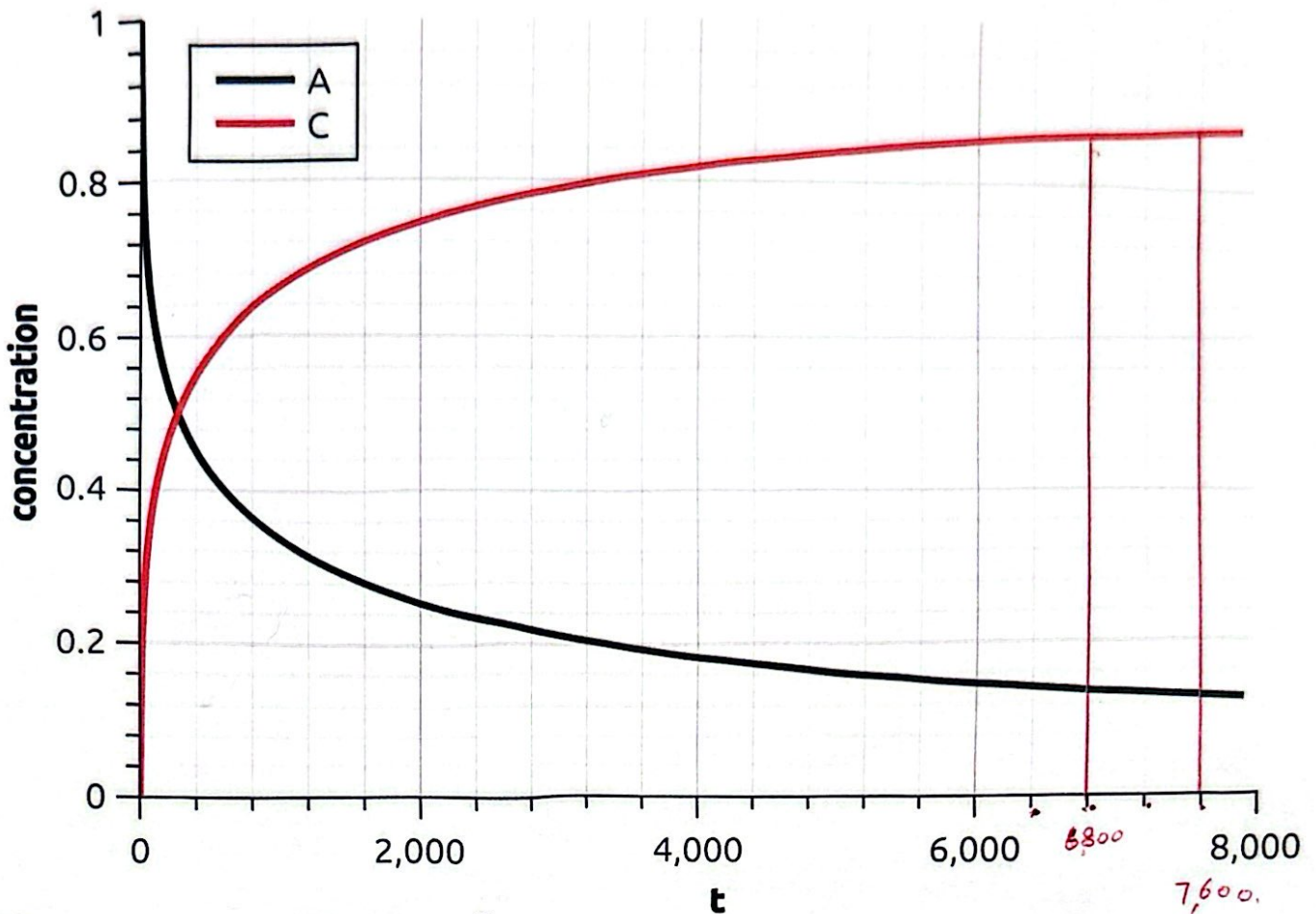
$[\tilde{A}]$ does not change much.

$\lambda_3 = 109.542$

$$V_3 = \begin{bmatrix} -1.210 \times 10^{-6} \\ -9.947 \times 10^{-1} \\ 1.028 \times 10^{-1} \end{bmatrix}$$

$$V_3^{-1} = [0.000, 0.000, 9.726]$$

Concentration of Species A and C vs t



③

$$t = 6,800 \quad \lambda_1 = 0, \quad v_1 = \begin{bmatrix} 8.458 \times 10^{-1} \\ 0.5335 \times 10^{-1} \\ 0 \end{bmatrix}$$

$$v_1^{-1} = [1.182, 8.679 \times 10^{-6}, 7.497]$$

$$\lambda_2 = -8.64 \times 10^{-3}$$

$$v_2 = \begin{bmatrix} -7.341 \times 10^{-6} \\ 1.000 \\ 0 \end{bmatrix}$$

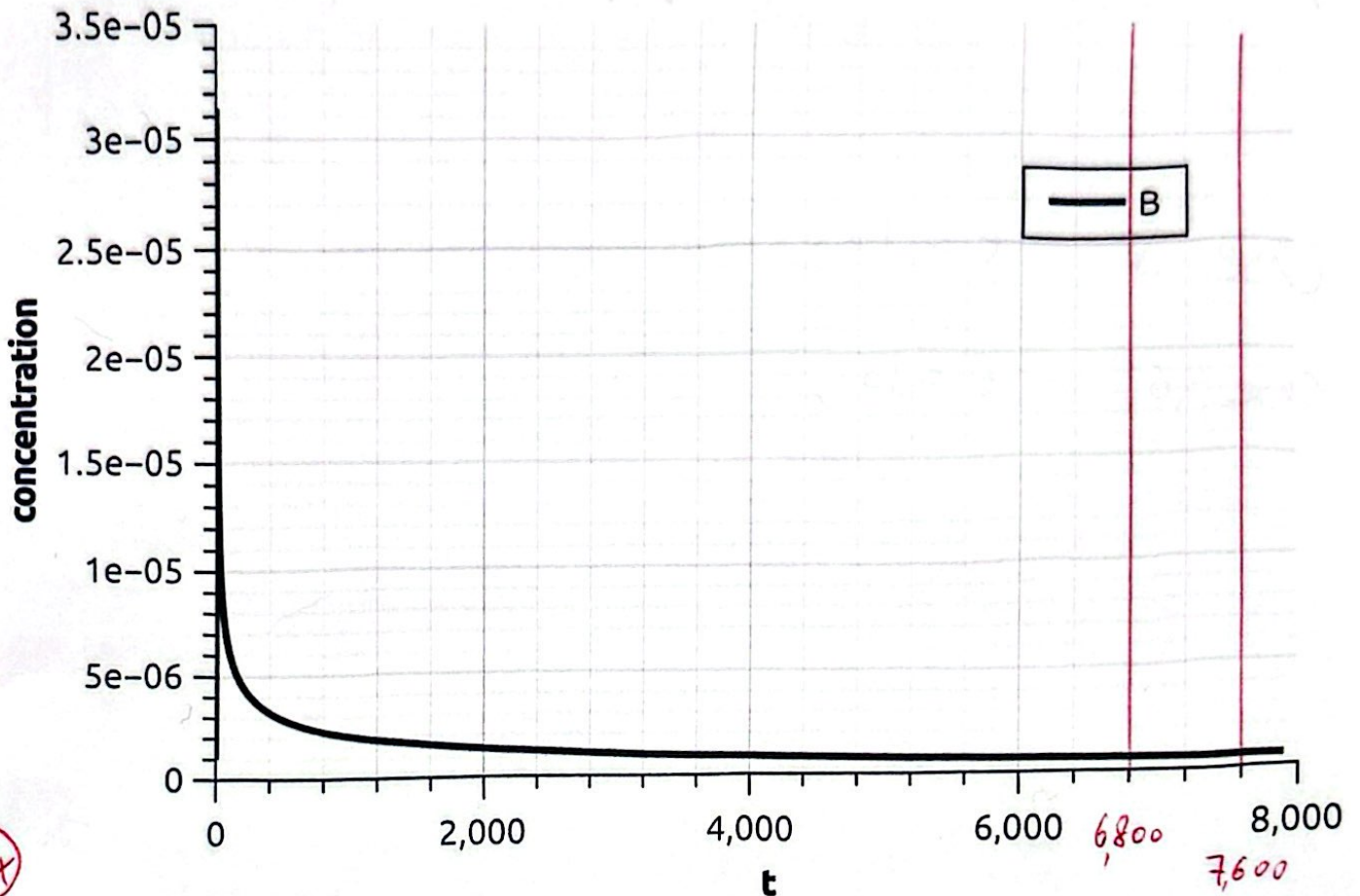
$$v_2^{-1} = [-6.308, 9.999 \times 10^{-1}, 3.473 \times 10^{-3}]$$

Concentration of σ does not change that much.

$$\lambda_3 = 3.473 \times 10^{-5} \quad v_3 = \begin{bmatrix} -8.382 \times 10^{-1} \\ -5.292 \times 10^{-1} \\ 1.322 \times 10^{-1} \end{bmatrix}$$

$$v_3^{-1} = [0.000, 0.000, 7.565]$$

Concentration of Species B vs t



(4)

$$t = 7,600 \quad \lambda_1 = 0 \quad v_1 = \begin{bmatrix} 8.635 \times 10^{-1} \\ 5.044 \times 10^{-1} \\ 0.000 \end{bmatrix} \quad v_1' = [1.158 \quad 9.089 \times 10^{-6} \quad 7.931]$$

$$\lambda_2 = -8.726 \times 10^3 \quad v_2 = \begin{bmatrix} -7.848 \times 10^{-6} \\ 1.000 \\ 0.000 \end{bmatrix} \quad v_2' = [-5.841 \times 10^{-1}, 9.999 \times 10^{-1}, 3.438 \times 10^{-3}]$$

$$\lambda_3 = 3.438 \times 10^{-5} \quad v_3 = \begin{bmatrix} -8.565 \times 10^{-1} \\ -5.007 \times 10^{-1} \\ 1.251 \times 10^{-1} \end{bmatrix} \quad v_3' = [0.000, 0.000, 7.995]$$

Something wrong with scaling: early stage: $[\tilde{A}]$ is in steady state

late stage: $[\tilde{B}]$ is in steady state.